- Project Title: Blood Bank Management System
- Tech Stack: MERN (MongoDB, Express, React, Node.js)
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Project overview

- **Abstract**: A full-stack solution to connect blood donors with recipients in real-time.
- **The Problem:** Traditional systems are manual, slow, and lack live updates on blood availability.
- **The Goal:** To automate inventory tracking and provide an instant search for life-saving blood groups.

System Analysis & Proposed Solution

- **Existing System:** Manual registers, fragmented data, and no remote access.
- **Proposed MERN Solution:Real-time Data: Live updates across the platform.**
- **Accessibility:** Cloud-based donor search from any device.
- **Security:** Protected donor information via JWT authentication.

MERN Stack Architecture

1 Frontend (React.js): Interactive dashboards using Hooks and Axios.

2 Backend (Node.js & Express): Secure RESTful APIs and business logic.

3 Database (MongoDB): Flexible NoSQL storage for donors and stock.

4 Middleware (Mongoose): Schema-based modeling for data consistency.

Key Modules & Functional Features

- **Donor Module:** Secure registration, profile management, and donation history.
- **Search/Seeker Module:** Filter blood by group (A+, O-, etc.) and check availability.
- **Admin Dashboard:** Control center for blood inventory, donor verification, and request approvals.

System Implementation & Testing

Implementation: Developed in VS Code; used Postman for API testing.

Unit Testing: Verified individual React components and Express routes.

Integration Testing: Ensured seamless "Form-to-Database" flow.

Security: Bcrypt.js used for password encryption.

Conclusion & Future Enhancements

Conclusion: The MERN stack provides a high-performance, scalable environment for critical healthcare data.

Future Scope:

GPS Integration: Locate donors via Google Maps API.

Notifications: Automated SMS/Email alerts for urgent needs.

Mobile App: Conversion to a cross-platform app using React Native.

THANK YOU

By Vignesh V